The pivotal role of metaphor -

word order routinization from proto- to modern language

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This paper argues that metaphorical expressions had a significant role to play in the evolution of modern-language grammar. We have outlined elsewhere how metaphor forces argument overtness (e.g., in *to arrive at the conclusion*, the metaphor-driving argument *conclusion* is always expressed in natural language use, lexically or pronominally, in contrast with a droppable literal goal as in *to arrive (at the hotel)*). In this paper, we show how the obligatory overtness of metaphor-driving arguments propels another hallmark of modern language, routinised word order. Whether or not word order is more rigid as in English, or more flexible as in most other languages, every language shows certain ordering defaults. We outline a scenario of how these defaults emerged as a by-product of the rise of metaphorical constructions in early stages of language evolution.

1. Introduction

This paper outlines the central role of metaphor in the development of modern-language syntax. We have shown elsewhere that metaphor forces argument overtness, thus having a significant - while previously almost entirely overlooked - impact on which arguments must be overt and which may be dropped, across languages and parts of speech. In this paper we build on this insight, arguing that not only argument expression, but also argument placement and word order rules at large are driven by the use of functor-argument metaphors. In the absence of direct evidence from proto-language, we show the effect of metaphor-driving arguments on the regularization of word order with evidence from the history of Indo-Aryan languages. We propose complementing this historical case study with computational simulations of ways in which ordering biases associated with metaphor-driving arguments lead to word order regularization.

2. Word order routinization in language evolution

Despite the still commonly used term "free word order", it is well-established that most and probably all languages show ordering defaults. Thus, even though few languages come with the syntactic rigidity of English, all languages make use of ordering norms both on clause level and on lower phrasal levels (Reinöhl, 2020). The shift in some of the literature to speak of "flexible word order" instead of "free word order" reflects this insight. Clause-level ordering tends to cater to semantic or pragmatic prominences (Himmelmann & Primus 2015; Ellison, 2024), such as agent-first order (Riesberg et al., 2019), topic-first order (Li & Thompson, 1976) or most-newsworthy-first order (Mithun, 1987). Phrase-level ordering is tightly connected to the grammaticalization of function words, which may turn flexible into rigid ordering. For instance, while early Latin lacked rigid ordering within nominal expressions, it sedimented as a by-product of the development of definite articles in Romance languages (Himmelmann, 1997; van de Velde, 2010, on the history of Dutch). In Indo-Aryan, nominal expressions showed significant flexibility in early Sanskrit, but, as a byproduct of the grammaticalizations of postpositions, developed the rigid right-headed defaults seen in modern Indo-Aryan (Reinöhl, 2016). Whether showing a certain degree of flexibility or rigidity, however, all of these cases attest to some ordering regularity (see also Schultze-Berndt & Simard, 2012; Louagie & Verstraete, 2016, for select Australian languages).

At the same time, we can plausibly assume that (flexibly or rigidly) ordered multi-word expressions were not always available in the history of human language, as this required multi-word utterances to be used in the first place. At the stage of human protolanguage when multi-word utterances first emerged, we hold that ordering cannot yet have settled on the defaults found in modern human languages - if only because multi-word utterances were new and rare and were not yet entrenched to a degree that could give rise to defaults. It is this phase that we are addressing in this paper: How did the word order defaults - characteristic of any modern human language - arise in the first place?

Note that this paper does not depend on a resolution of the holistic vs. analytic debate, i.e. the question of whether the earliest utterances consisted of single "words" that had complex, holistic speech-act-like meanings ("There comes a predator!", "The food is good." Etc.) or isolated, analytic meanings known from e.g. modern European languages where a word only encodes a part of a message, e.g. "come" or "food". Our discussion is neutral with regard to this debate because

our argument takes off only once we are dealing with utterances consisting of two or more words, which together form a single message.[1]

3. Metaphor-driven argument overtness

Before we continue our argument regarding the emergence of word order defaults, we now turn to another precondition: the overt expression of arguments in metaphorical functor-argument combinations (where functor stands for any term that has at least one slot for an argument). Notably, even numerous modern languages allow, under particular conditions, the covert encoding of arguments. In some languages, this covert encoding is seemingly salvaged by the existence of inflectional or pronominal morphological marking attaching to the verb or verb complex. The label "radical pro drop", however, captures cases where an argument is dropped without any overt cross-referencing. An example of this is the Japanese statement *daisuki desu*, literally '_ love(s) _', which is a satisfactory statement in an appropriate context, even though it specifies neither a subject nor an object.

By contrast, English is well-known for its particularly strong demands for arguments expression, even when these are semantically empty (cf. *it* in *it rains*). But English, too, allows for covert expression of (semantic) arguments, such as those expressed by prepositional phrases with appropriate priming (*They arrived.* vs *They arrived at the hotel.*). Whether language-specific restrictions on argument expressions are at the tighter (English) or looser (Japanese) end of the spectrum, Reinöhl & Ellison (2024) show that there are certain conditions under which arguments must be expressed in every language and across part-of-speech systems (see also Ellison & Reinöhl, 2022). Compare examples (1) and (2). In this pair, we contrast literal and metaphorical uses of the functor *arrive*. Someone physically arrives somewhere in (1), but no spatial movement is involved in the "arriving" event in (2). The argument *conclusion* clashes with the selectional restriction on arguments imposed by the literal meaning of the functor and so forces metaphorical re-interpretation. This re-interpretation, however, only becomes available with the overt pronominal resumption of the clashing argument. Reinöhl & Ellison (2024) present corpus evidence from English (from

[1] The incorporating morphology found in some modern languages is a later development, a compression of syntactic building blocks into morphologically complex words (see Mithun, under review). Also note that proponents of a synthetic origin (e.g. Wray, 1998) imagine the piecemeal splitting-up of the early holistic message-words into several words, where they essentially reach the stage that is proposed as the starting point in analytic-first accounts (e.g., Tallerman 2007).

a corpus including written texts and spontaneous speechsimu), Indo-Aryan and Vera'a, as well as from a sentence-completion experiment in English, that this generalization holds near-absolutely[2].

(1) *Everybody else arrived at **the hotel** on Monday.*
*John **arrived (at it)** on Tuesday.*

(2) *Everybody else arrived at **that conclusion** on Monday.*
*John **arrived at it** on Tuesday.*

4. Metaphor and word order routinization

We proposed elsewhere that metaphor is not only a driving force of language evolution in terms of its potential to expand expressivity (see also Smith & Höfler, 2015 & 2017), but also in terms of its direct impact on morpho-syntactic structure, including word order. As Ellison & Reinöhl (2022) put it:

"Natural discourse can very well do without much overt expression: since the same participants tend to be tracked through a stretch of discourse, they can be economically expressed with zero anaphors (e.g. Du Bois, 1987). Only the use of metaphor forces argument overtness, even under conditions of maximum activation. [...] With enough material expressed overtly often enough, the stage is set for conventionalization." (2022, no page)

Additional evidence supports this proposal. A constraining factor, however, is that any linguistic data comes from a modern language in an evolutionary sense - including historically attested data such as that of Vedic Sanskrit. In other words, what we spell out here is a conceptual argument supported, but not proved or provable by evidence showing the process in action. We return to this point in section 5.

As outlined above, metaphorical readings arise from a clash in literal meanings of functor and argument. Even when mentioned very recently (such as, as the last word of the last utterance), the metaphor-driving argument will need to be overtly resumed. With highly activated arguments, pronominal resumption is the default. Our question is now: Where in the utterance is this metaphor-driving pronominal argument going to be placed? As shown above, the argument in corresponding literal expressions (e.g. ex. (1)) does not have to be expressed at all. Since the overt expression of the metaphor-driving argument thus corresponds to a literal zero, we assume that the former's placement in the clause is a default position of

[2] Note that the sentence-completion experiment was done through selection options from a menu, and none of the options were zero-length – e.g. "waves" vs "waves of it" – to avoid a potential bias from participants avoiding empty responses.

a sort; after all, it is only expressed for the semantic metaphor constraint, while it could have remained unexpressed if only information-structural constraints were active. We thus argue that a pronominally realized, metaphor-driving argument lands in what can be conceived as a default slot - free of the information-structural and pragmatic constraints otherwise impacting placement. Here are examples from different historical varieties of Indo-Aryan (ex. 5 includes context with the previously mentioned argument ('gamblers'), left out for reasons of space in ex. 6).

(5) Apabhramsha (Late Middle Indo-Aryan)

tahiṃ	*jāivi*	*ṭiṇṭahiṃ*	*gau*	*turantu /*
DEM.LOC.SG	go.CONV	gambling-house.LOC.SG	go.PPP.NOM.SG.M	in_haste

jūvārahaṃ	*sayalahaṃ*	*maṇu*	*harantu /*
gambler.GEN.PL	all.GEN.PL	mind.ACC.SG	grab.PPA.NOM.SG.M

tahaṃ	***majjhi***	*ṇiviṭṭhau*	*so*
DEM.GEN.PL	madhye	settle_down.PPP.NOM.SG.M	DEM.NOM.SG.M

sahei /
shine.3SG

chaṇaindaho	*līlā*	*ṇam*	*vahei*
full_moon.GEN.SG	beauty.ACC.SG	so	bear.3SG

'Reaching there, (the king) quickly went to the gambling house attracting the mind of all the gamblers. Sitting in **their middle** he shines as if bearing the beauty of the full moon.' (KA 8.15.3, cp. Jain)

(6) Vedic Sanskrit

Tasya	***madhye***	*nividaṃ*	*dadhāti*
DEM.GEN.SG.M/N	middle.LOC.SG.N	short_invocation.ACC.SG.F	put.PRS.3SG

'In the middle of it (i.e. of a recitation), he places a short invocation' (Śā̃ 15.3a)

While not absolute, the ordering regularity with pronominal arguments contrasts with the greater ordering diversity found with nominal arguments (Reinöhl 2016, esp. Tables 6, 7, 8). For example, whereas 18 out of 19 cases of pronominal arguments show the adjacent order *GEN functor* in Apabhramsha (Late Middle Indo-Aryan, ca. 8th-13th cent. AD), with no case of discontinuity attested already since Pali (early Middle Indo-Aryan, ca. 500 BC). By contrast, cases with nominal possessors show greater variation in adjacency (e.g., 22 out of 33 cases for GEN *madhye*) and also still allow discontinuity (8 cases out of a total of 38 cases in Pali, 2 cases out of 35 in Apabhramsha).[3]

In other words, the fixed postpositional order known from modern Indo-Aryan languages is already preshadowed in combinations of the grammaticalizing

[3] Reinöhl (2016: 139-144) additionally discusses the tendency of the pronominal argument to not only precede the functor, but also to appear in clause-initial position. This can be linked to the clause-linkage use of demonstratives.

functors with pronominal arguments (e.g., PRO *madhye* 'in the middle of X' will, over time, turn into e.g. Hindi PRO *mẽ* 'in X'). In fact, it is only the order argument-functor which undergoes grammaticalization, as illustrated in non-grammaticalized interpretations of the rare early Hindi examples of the reverse order (see Reinöhl, 2016: 167; e.g. 'in the middle of the confluence [of Ganges, Yamuna and Sarasvati]' rather than 'in the confluence').

5. Outlook - simulating word order evolution

While the evidence reviewed in section 4 supports our argument, it cannot be seen as offering proof in view of the fact that even the very earliest strata of Sanskrit are far removed from the roots of human language evolution. While the constructions in question thus do show the development of (more rigid) word order conventions, the surrounding linguistic system possesses, in other grammatical domains, already some other such conventions. For example, Indo-Aryan has always been predominantly verb-final from its earliest attestations (Delbrück 1878, 2009 [1888]; Reinöhl 2016: 119-145). In future work, this empirical gap may be approached from another angle, with computer simulations recreating the asymmetry of the impact of argument optionality on a Bayesian model of interpretation under literal functor-argument combinations, and obligatory argument overtness under metaphorical conditions. These conditions could feed iterations of speaker-to-speaker and generational transfer, and potentially reveal whether or not (metaphor-driving) pronominal arguments indeed lead the way in word order regularization.

6. Conclusion

We have argued in this paper that metaphorical functor-argument constructions do not only require the overtness of the metaphor-driving argument, but also drive word order regularization. This effect is linked to the obligatory pronominal resumption of activated metaphor-driving arguments, a phenomenon that we have shown elsewhere to plausibly apply universally and across part-of-speech categories. Pronominal resumption in cases of highly active, but metaphor-driving arguments, contrasts with the oft-found zero expression of corresponding literal arguments. This correspondence, we argue, means that pronominal metaphor-driving arguments are placed in what can be considered a default position, as their overtness is only forced by semantic, not information-structural requirements. This default placement, in turn, over time sediments with an increase in metaphorical functor-argument constructions, giving rise to the modern-language word order defaults found across languages, from near-absolute rigidity as in English, to more flexibly, but still default-based, orderings in other languages.

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